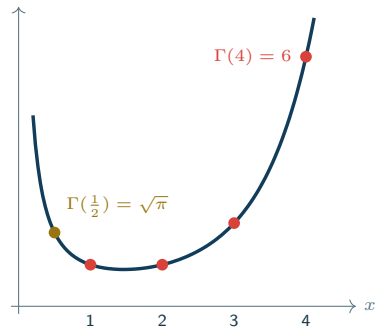
FUNCTIONS LADDER 9 OF 10

The gamma function and the beta function

The gamma function continues the factorial to every complex number, and the beta function is a ratio of gamma values.

1 / THE RULE AND THE PICTURE

The gamma function interpolates the factorials, with $\Gamma(n) = (n-1)!$



The curve passes through the factorials $\Gamma(1) = 1$, $\Gamma(2) = 1$, $\Gamma(3) = 2$, $\Gamma(4) = 6$, and between them it is smooth, with a dip to a minimum 0.886 near $x = 1.46$.

The defining integral and the functional equation obtained by integration by parts are

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt, \qquad \Gamma(z+1) = z\,\Gamma(z)$$

2 / THE RECURRENCE AND HALF-INTEGERS

Integrating by parts shifts z by one, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ is the Gaussian integral

Writing $\Gamma(z+1) = \int_0^\infty t^z e^{-t} dt$ and integrating by parts gives $\left[-t^z e^{-t}\right]_0^\infty + z\int_0^\infty t^{z-1} e^{-t} dt = z\Gamma(z)$, so $\Gamma(n) = (n-1)!$ by induction. The substitution $t = u^2$ turns $\Gamma(\frac{1}{2})$ into $2\int_0^\infty e^{-u^2} du = \sqrt{\pi}$, and the recurrence then gives every half-integer: $\Gamma(\frac{3}{2}) = \frac{1}{2}\sqrt{\pi}$, $\Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi}$. Running the recurrence backwards, $\Gamma(z) = \Gamma(z+1)/z$ extends the function to negative arguments, with poles at $0, -1, -2, \ldots$.

3 / SOME VALUES

The reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$ ties z to $1-z$

argument	exact	value	argument	value
$\Gamma(\frac{1}{2})$	$\sqrt{\pi}$	1.7725	$\Gamma(\frac{7}{2})$	$\frac{15}{8}\sqrt{\pi} = 3.3234$
$\Gamma(\frac{3}{2})$	$\frac{1}{2}\sqrt{\pi}$	0.8862	$\Gamma(-\frac{1}{2})$	$-2\sqrt{\pi} = -3.5449$
$\Gamma(\frac{5}{2})$	$\frac{3}{4}\sqrt{\pi}$	1.3293	$\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})$	$\pi\sqrt{2} = 4.4429$

At $z = \frac{1}{4}$ the reflection formula gives $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi/\sin\frac{\pi}{4} = \pi\sqrt{2} = 4.4429$, matching 3.6256×1.2254, and the unit ball in n dimensions has volume $\pi^{n/2}/\Gamma(\frac{n}{2}+1)$: π for $n = 2$, 4.189 for $n = 3$, 5.264 for $n = 5$ and 2.550 for $n = 10$.

4 / THE BETA FUNCTION / THE POCHHAMMER SYMBOL

The beta function is a ratio of gamma values

$$B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

For integers $B(2,3) = \dfrac{1!\,2!}{4!} = \frac{1}{12}$, and for $a = b = \frac{1}{2}$ the integral is $\int_0^1 \frac{dt}{\sqrt{t(1-t)}} = \pi = \dfrac{\Gamma(\frac{1}{2})^2}{\Gamma(1)}$. The ratio $(a)_n = \dfrac{\Gamma(a+n)}{\Gamma(a)} = a(a+1)\cdots(a+n-1)$, the rising factorial or Pochhammer symbol, satisfies $(1)_n = n!$ and $(\frac{1}{2})_3 = \frac{1}{2}\cdot\frac{3}{2}\cdot\frac{5}{2} = \frac{15}{8}$, and it is the building block of hypergeometric series.

5 / STIRLING'S APPROXIMATION

Large factorials are $n! \approx \sqrt{2\pi n}\,(n/e)^n$

For $n = 10$ the formula gives $3\,598\,696$ against $10! = 3\,628\,800$, an error of 0.83%, and for $n = 20$ it gives 2.4228×10^{18} against 2.4329×10^{18}, an error of 0.42%; the relative error is about $\frac{1}{12n}$.

The same formula holds for $\Gamma(z+1)$ at real z, and it is the reason entropy per particle is $\ln N! \approx N \ln N - N$.

ASIDE / A DEFINITION BY PROPERTIES

The only log-convex function with $f(1) = 1$ and $f(x+1) = xf(x)$ is Γ, which is what makes it the right continuation of the factorial.

LINKS TO OTHER SHEETS

The Gaussian integral also gives the normal distribution in *The normal distribution and the central limit theorem*; the previous sheet is *Taylor series* and the last is *Hypergeometric functions*.

Convention. Γ is the gamma function; B the beta function; $(a)_n$ the rising factorial; π appears through $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.